

An unexpected source of invertebrate community recovery in intermittent streams from a humid continental climate

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Abstract

1. Intermittent streams are naturally dominant landscape features of Mediterranean and arid regions, but also occur more and more in humid climates, such as in the Czech Republic. Organism abilities to cope with drying (i.e. resistance forms) have been quantified in naturally intermittent streams from Mediterranean and arid regions, in which long-term flow intermittence patterns have promoted physiological adaptations. In contrast, the capacity of aquatic communities to persist without requiring resistance adaptations under recent intermittent flow regimes in more humid climates is virtually unexplored, along with its contribution to community recovery upon rewetting.
2. Here, we addressed the ability of aquatic invertebrates devoid of specific desiccation-resistance forms (e.g. cysts or eggs) to cope with climate change-induced flow intermittence from rivers in continental climate. Owing to the high relative air humidity, the importance of riparian cover and the short duration of drying events, we expected taxonomically and functionally rich communities to persist in the dry streambed during phases with no surface water and to contribute to community recovery upon flow resumption.
3. To explore these ideas, we collected invertebrate communities in the riverbeds before, during and after drying from 10 intermittent rivers in the Czech Republic. We quantified the pool of organisms remaining alive in the dry riverbeds and determined its taxonomic and functional contribution to the recovery of benthic communities upon rewetting.
4. Of the organisms collected, 83%, belonging to 22 taxa, were able to survive during the dry phase without producing desiccation-resistance forms. This pool of organisms contributed substantially both taxonomically and functionally to the quick recovery of benthic communities. The exponential decrease in taxonomic and functional richness implies an initial quick loss of sensitive taxa, while a pool of less sensitive taxa persisted long in the dry riverbeds, hence contributing to the fast community recovery upon rewetting.
5. Survival rates of active aquatic macroinvertebrates devoid of specific desiccation-resistance adaptations have rarely been quantified. Probably promoted by benign environmental conditions prevailing on dry riverbeds in this study, their survival

demonstrated the need to protect communities in dry riverbed and their adjacent riparian environments so that assemblages can recover rapidly from short periods without river flow. Riparian forest management along intermittent streams from continental climates could help conserve the value of dry streambeds as a source of resilience for riverine communities.

KEYWORDS

drying, global change, similarity, temporary rivers, tolerance

1 | INTRODUCTION

Intermittent streams are naturally prevalent in river networks all over the globe (Datry et al., 2014; Larned, Datry, Arscott, & Tockner, 2010). While they are a dominant landscape feature of Mediterranean and arid regions, they also occur abundantly in more humid climates (Buttle et al., 2012; Stubbington et al., 2017). For example, a large proportion of western, humid France is composed of intermittent streams (Snelder et al., 2013). In response to climate change and increased needs for water, many perennial rivers are becoming intermittent (Döll & Schmied, 2012; Gudmundsson, Seneviratne, & Zhang, 2017; Laaha et al., 2016). This is particularly true in areas where intermittent rivers were uncommon in the past such as the Czech Republic (CR). Prone to a continental, humid climate (Peel, Finlayson, & McMahon, 2007), the stream network of CR was dominantly composed of perennial waters until recently when the proportion of intermittent streams has grown considerably: in the past decade, many streams have started to dry for the very first time and today, 50% of stream length in the country is at risk of drying (Benčoková, Krám, & Hruška, 2011; Van Lanen et al., 2016; Zahrádková et al., 2015).

Naturally intermittent streams are characterised by a high variability of habitats, alternating in time and space between aquatic lotic, lentic to terrestrial conditions which accommodate rich and diverse biotic communities (Datry et al., 2014). This dynamism has triggered physiological adaptations for many aquatic species to cope with flow cessation and drying events (Bogan et al., 2017; Strachan, Chester, & Robson, 2015; Williams, 2006). Some species can, for example, persist weeks or even months *in-situ* during drying as desiccation-resistance forms in dry streambed sediments (Storey & Quinn, 2013; Stubbington & Datry, 2013). Others are able to escape drying reaches and quickly recolonise upon flow resumption from nearby perennial refuges, such as perennial reaches in the network (Bogan, Boersma, & Lytle, 2013) or the underlying hyporheic zone (Vander Vorste, Malard, & Datry, 2016). Others may persist in moist interstices in dry streambeds (Boulton, Stanley, Fisher, & Lake, 1992; Strachan et al., 2015), although the importance of this pool of taxa to contribute to community recovery upon rewetting has not been quantified to date. Invertebrate resistance strategies to drying have been well quantified in naturally intermittent streams from Mediterranean and arid regions (Bogan et al., 2017;

Boulton et al., 1992; Robson, Chester, & Austin, 2011; Stubbington & Datry, 2013), whereas the capacity of aquatic communities to persist under novel intermittent flow regimes is underexplored (Datry, Bonada, & Boulton, 2017; Datry, Vander Vorste, et al., 2017). In humid areas where most rivers and streams have been historically perennial, the resilience of aquatic communities to on-going drying trends is unclear. In the CR, small streams (up to 4th Strahler order) dominate (90% of total stream length) and have become prone to frequent drying events in the last decade as a result of climate change and increased water abstraction for agriculture (Trnka et al., 2016; Van Lanen et al., 2016; Zahrádková et al., 2015). How resilient are the communities from these *novel* ecosystems (*sensu* Hobbs et al., 2014) affected by anthropogenic flow intermittence is an open question. Drying events in humid areas are unpredictable and rather short, typically <1 month (Pařil et al., 2019; Řezníčková, Pařil, & Zahrádková, 2007; Řezníčková, Soldán, Pařil, & Zahrádková, 2010). The lack of adaptation of species that have not experienced long-term flow intermittence in the past could make them highly vulnerable to anthropogenic flow intermittence induced by direct (e.g. water abstraction, channel modification) and indirect (e.g. climate change) causes. Conversely, the high relative humidity, promoted by the dense riparian cover and short dry periods could allow species to tolerate desiccation and persist despite anthropogenic flow intermittence.

In this study, we addressed the ability of aquatic invertebrates from continental humid climates to cope with direct and indirect anthropogenically-induced flow intermittence. We sampled invertebrate communities before, during and after several drying events, in 10 near-natural streams of the CR with minimal direct human impact. Owing to the high relative air humidity, substantial riparian cover and the short duration of drying events, we expected taxonomically and functionally rich communities to persist in the dry streambeds during phases with no surface water without requiring desiccation-resistance forms. We also expected predictable patterns of β -diversity and its components of these communities based on recent studies (Schriever et al., 2015; Vander Vorste et al., 2016). We first predicted both taxonomic and functional richness of communities in dry streambeds to decrease with increasing duration of the dry phase to increase again upon flow resumption. This is because dry phase duration is known to be a primary control of invertebrate communities (Stubbington & Datry, 2013), while benthic

communities recover gradually upon flow resumption (Leigh et al., 2016). Secondly, we predicted a dominance of nestedness for functional β -diversity and a dominance of turnover for the taxonomic β -diversity of dry streambed sediments. This is due to the functional redundancy of intermittent stream communities, in which different taxa share identical functional trait combinations (Datry, Bonada, & Boulton, 2017; Datry, Vander Vorste, et al., 2017; Schriever et al., 2015; Vander Vorste et al., 2016). Thirdly, we predicted that a substantial proportion of river communities will be supported upon rewetting by the pool of tolerant species, both structurally and functionally, indicating that this species pool could promote the recovery of communities upon rewetting.

2 | METHODS

2.1 | Study sites

Ten river reaches were selected throughout the CR (range of latitude 48.8°–50.5°, range of longitude 13.9°–17.6°, Figure 1) and within the Dfb Köppen class (Peel et al., 2007), indicating a humid continental climate. The selection of the reaches was based on preliminary streambed temperature and drying observations through photo-traps and temperature loggers, and was intended to span a wide gradient of drying duration. Selected reaches ranged in altitude from 250 to 480 m, in a region receiving 640 mm of

annual rainfall with a mean temperature of 7.9°C (Tolasz, Míková, Valeriánová, & Voženílek, 2007) and placed proportionally on siliceous (five sites) and calcareous (five sites) bedrock. For these 2–4th Strahler order streams, distance to the source ranged from 2.7 to 10.9 km (mean 5.8 km). Reaches were all pristine, without influence of contamination, channel modification, or water abstraction, while the upstream catchments were dominated by forest (55%), extensive agriculture (43%) and <5% urbanisation (CORINE Land Cover 2017, Table S1).

2.2 | Study design

Our study design was as follows: first, we sampled invertebrate communities before, during and after several drying events in the 10 intermittent streams of the CR. Second, we explored the taxonomic and functional diversity independently for each of these three phases and related them with drying and flowing patterns. Third, we quantified the similarities among the three phases and particularly explored that of post-drying samples with during-drying samples using presence–absence data to quantify the potential contribution of communities of the dry period to the recovery of benthic communities (both from taxonomic and functional perspectives) upon rewetting. This design was based on (Datry, Corti, & Philippe, 2012; Datry, Bonada, & Boulton, 2017; Datry, Vander Vorste, et al., 2017).

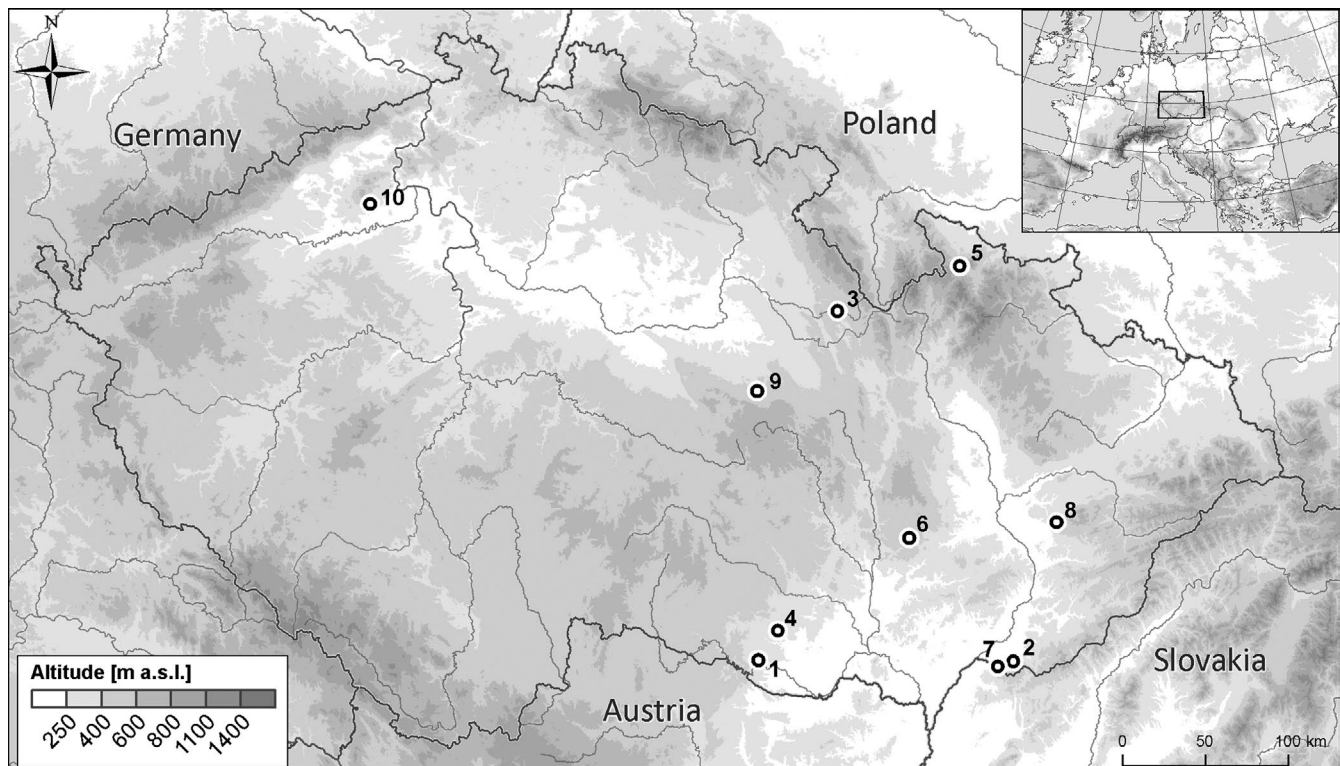


FIGURE 1 Map showing the sampling sites within the 10 intermittent streams of the Czech Republic. Stream name and sampling site: 1. Granický p.—Znojmo, 2. Jarkovec—Lucina, 3. Kamenická—Zámek, 4. Krepická—Horní Dunajovice, 5. Obloučník—Vapenna, 6. Rakovec—Bukovinka, 7. Sudoměřský p.—Radejov, 8. Zhrta—Jankovice, 9. Zejbro—Dolívka, 10. Zejdlík—Třebívlice. Site characteristics are detailed in Table S1

2.3 | Drying duration and frequency

To quantify the drying phase characteristics, one temperature logger (HOBO Onset Pro v2 U22-001, Onset Computer Corporation), one water level logger (Solinst Levellogger Edge, Solinst Canada Ltd.), and one photo-trap (Acorn 5310MG, Zhuhai Ltl Acorn Electronics Co., Ltd.) were installed at each reach. Water level loggers were placed in the deepest section of the streambed to account for the persistence of surface water in residual pools, while temperature loggers were placed in shallower section of the streambed to detect flow cessation and record temperatures prevailing on dry riverbeds. The photo-traps were installed in trees in the adjacent riparian zone (at 3 m height) and were facing a graduated scale to measure the water level in the streambed, where temperature and water level loggers were placed. Data were collected every 30 min on loggers and each 4 hr on photo-trap.

Based on all collected data, the extent of surface water drying with one-day resolution was recorded for all sampled streambeds. The number of days with no flow prior to sampling date in sampling year and number of days after rewetting prior to sampling date were used in subsequent analyses as explanatory variables.

2.4 | Macroinvertebrate sampling

We sampled macroinvertebrates from each reach during flowing phases before and after dry periods and during the dry period, for 3 years: 2013 (six reaches), 2014 (seven reaches), and 2015 (five reaches). Altogether, 54 samples were collected from the 10 stream reaches.

2.4.1 | During dry phase

Dry sediments (DRY) were sampled during summer between 26 July and 28 August, 1–152 days following drying phase onset. At each reach, an area of 20 × 20 cm was randomly chosen in the lowest point of one riffle and one pool and pooled together. At each area, about 5 L of sediments were collected to a depth of 12.5 cm with a shovel, placed in plastic bags and transported to laboratory in cool boxes. In the laboratory, samples were processed within 24 hr. Samples were washed through 500-µm nets and macroinvertebrates sorted under a stereomicroscope with 50× magnification. Dead and alive organisms were distinguished and separated based on apparent living signs, including movement of organisms. Stages <0.5 mm, such as eggs or cysts were disregarded and not considered here, as our aim was not to focus on desiccation-resistance stages, largely explored by other studies (Storey & Quinn, 2013; Stubbington & Datry, 2013).

The key environmental variables known to influence the survival of organisms in intermittent rivers (Strachan, Chester, & Robson, 2016; Stubbington & Datry, 2013) were quantified. Basic characteristics of dried bed sediment were identified by granulometric analysis on consecutive sieves (mesh sizes [cm]: 6.3, 3.15, 1.6, 0.8, 0.4, 0.2, 0.1, 0.05, 0.025, 0.0125, 0.0063) and each fraction was weighed. Sediment moisture was measured for 70 randomly selected samples by collecting during the dry phase 1 L of sediments in the first

12.5 cm of the streambeds and in the vicinity (<50 cm) of invertebrate sampling areas. Stored in sealed plastic bags, the samples were brought to the laboratory within 24 hr, were weighed wet, dried at 105°C to a constant mass and reweighed. Moisture was determined from differences between wet and dry mass.

2.4.2 | During flowing phase

Macroinvertebrates were sampled before and after the dry periods using the Czech national standardised protocol *PERLA* (Kokeš et al., 2006). It is based on a proportional multihabitat 3-min kick sampling using a pond net (frame 25 × 25 cm; mesh size 500 µm; 1-m long mesh). Samples were pre-sorted in the field by experienced limnologists for 20 min to ensure detection of rare taxa. Pre-picked macroinvertebrates and remaining samples were then separately preserved in 4% formaldehyde and further processed in the laboratory using stereomicroscopes.

Pre-dry period samples (PRE) were collected in spring between 28 March and 29 April, whereas post-dry period samples (POST) were collected in autumn between 27 September and 21 November. POST samples were collected typically 1 month upon flow resumption (mean = 38 days). Due to severe drought in 2015, two reaches were sampled only 2 and 3 days after flow resumption at the end of October to avoid sampling during winter when streams were frozen.

2.5 | Invertebrate identification and species traits

All macroinvertebrate taxa were identified to the lowest taxonomic resolution according to obligatory identification level stated in the Czech IS ARROW database (<http://hydro.chmi.cz/isarrow/>); (Czech Hydrometeorological Institute 2018) using current literature predominantly to species (71%) and genus (20%) levels (only 19% to family of higher taxonomic units) in PRE and POST samples (but due to a less precise level of identification in dry sediment, the whole dataset was adjusted according to DRY samples—see Data analysis).

The trait database assembled during the BIODROUGHT project (www.biodrought.eu), which contains information about 35 different species traits related to ecology, life history, physiology, and morphology of Central European freshwater macroinvertebrate taxa, was used for general characterisation of the dry streambed community. This database integrates information from >500 published articles and other resources (such as the database freshwaterecology.info and Tachet, Richoux, Bournard, & Usseglio-Polatera, 2010). Altogether, 12 biological traits with 53 trait categories, which have been shown to respond to flow intermittence (Aspin et al., 2019; Datry et al., 2014) were selected (Table 1). All traits are fuzzy-coded in range 0–1. Using 53 trait categories, the majority of taxa possessed at least slightly different combination of traits and therefore, almost every taxon represented different functional units. Because of that, we defined the functional diversity as the number of general functional groups within a sample and estimated the functional diversity following the approach presented in Aspin et al. (2019). We defined the trait profile groups, i.e. groups

TABLE 1 List of species traits used for cluster analysis to separate taxa into 18 trait profile groups

No.	Trait	Category	No.	Trait	Category
1.	Feeding	Specialist	8.	Locomotion	Swimming/scatting
		Generalist			Swimming/diving
2.	Temperature preference	Very cold	9.	Life cycle type	Burrowing/boring
		Cold			Sprawling/walking
		Moderate			(Semi)sessil
		Warm			Other
3.	Armouring	Eurytherm	10.	Reproduction	Semivoltine
		Soft			Univoltine
		Softly sclerotised			Bivoltine
		Sclerotised			Trivoltine
4.	Resistance to drying	Armoured	11.	Body flexibility	Multivoltine
		No drought resistance			Flexible
		Egg diapause			Ovovivipar
		Larvae diapause			Free isolated eggs
5.	Fecundity	Adult diapause	12.	Respiration	Cemented isolated eggs
		Unknown resistance type			Fixed clutches
		<100 eggs in hatch			Free clutches
		100–300 eggs in hatch			Clutches in vegetation
		300–500 eggs in hatch			Terrestrial clutches
		500–1,500 eggs in hatch			Asexual
6.	Dissemination strategy	1,500–3,000 eggs in hatch	11.	Body flexibility	Parasitic
		>3,000 eggs in hatch			None
		Aquatic passive			Low
		Aquatic active			High
7.	Dissemination potential	Aerial passive	12.	Respiration	Tegument
		Aerial active			Gill
		<10 m			Plastron
		10–100 m			Spiracle (aerial)
		100–1,000 m			Hydrostatic vesicle (aerial)
		>1,000 m			Tapping of air stores of aquatic plants
					Extension/excursion to surface

Note: Traits were extracted from the Biodrought database. Numerical values of traits except first categorical trait *feeding* vary between 0 and 1. Trait 11. Body flexibility is related to capability of substrate interstices penetration and increasing with higher flexibility. Trait 6. Dissemination strategy is related to dispersal mode which enable successful recolonisation of intermittent streams and increasing from poor *aquatic passive* to most efficient *aerial active* dispersal. Detailed definition of traits 1, 2, 4, 6, 7, 8, 9, 10, and 12 are available on freshwaterecology.info (Schmidt-Kloiber & Hering, 2015).

of functionally similar (or equal) species and used such groups as units for functional diversity quantification. To find most homogeneous groups of taxa according to their trait, the Gower dissimilarity matrix was calculated from species $\times$ traits matrix and Ward's hierarchical grouping method was used as clustering method (Aspin et al., 2019). The optimal number of the most homogenous (i.e. with lowest average cluster width) clusters has been defined using cluster width criterion using R package cluster (Maechler, Rousseeuw, Struyf, Hubert, & Hornik, 2017).

All species belonging to same cluster constitute the same functional unit. Two different matrices were therefore used for

subsequent analyses: species $\times$ samples matrix and functional units $\times$ samples matrix.

2.6 | Data analysis

First, accumulation curves were plotted against the number of samples to compare the sampling effort among the three type of samples (PRE, DRY, POST) and ensure similar sampling effort (Figure S1).

To ensure consistent taxonomic resolution among sample types (PRE, DRY, POST), we used the resolution of DRY samples, which contained small and sometimes indistinguishable juvenile

and partly damaged specimens. In general, the species level was reached for 57%, genus level for 38%, and family and higher level for 5% of all taxa. All singletons (taxa occurring only once in the whole dataset) and taxa observed exclusively in compared samples from DRY sediments (and therefore not contributing to flowing-phase community) were removed prior to subsequent analyses.

2.6.1 | Taxonomic and functional richness

Taxonomic (i.e. number of taxa) and functional richness (i.e. number of functional groups—for details see Bonada, Doledéc, & Statzner, 2007) and total abundance were compared among sample types using one-way analysis of variance (ANOVA). We used trait profile groups as functionally non-redundant units for functional diversity estimation and defined the functional diversity as number of functionally highly similar groups of taxa (see Aspin et al., 2019). To test our first prediction, the relationships between taxonomic and functional richness in DRY samples and the dry phase duration was explored by linear regression, as for the relationship between taxonomic and functional richness of POST samples and the duration since flow resumption. The differences in the relationships between duration of drying and both taxonomic and functional diversity were assessed using an analysis of covariance (ANCOVA, exploring the significance of the factor interaction). Similarly, the differences in the relationships between the duration of the flowing phase and both taxonomic and functional diversity were assessed using an ANCOVA. The normality of data was visually checked using quantile–quantile plots, the homogeneity of variances was tested using Bartlett tests and variables were log-transformed to meet homoscedasticity.

2.6.2 | Taxonomic and functional β -diversity partitioning

To explore the β -diversity of DRY samples, we calculated (1) turnover component, (2) nestedness component, and (3) total Bray–Curtis dissimilarity for both (taxonomic and functional) types of dry streambed communities following Baselga (2013), for all pairs of DRY samples within the same year. The three β -diversity measures were then compared between DRY and POST samples using non-parametric Kruskal–Wallis H tests.

2.6.3 | Dry streambed community contribution to flowing-phase communities

We calculated the potential contribution of DRY samples to community composition in POST samples during recovery in two different ways. We first compared the number of total, shared and unique taxa and functional units among DRY, PRE, and POST samples (abundance data were not used due to the discrepancy in sampling methods among DRY and PRE/POST samples). The ratio of the number of shared taxa by the total number of taxa in the

flowing phase community defined the contribution in terms of shared species. Similarly, the ratio of the number of shared functional units by the total number of functional units in the flowing phase community defined the contribution in terms of functional diversity. Secondly, we quantified taxonomic and functional community dissimilarity among sample types for each site using the Chao index calculated on presence–absence data, which is particularly suitable when sample size and taxonomic richness differ among groups (Anderson & Millar, 2004; Cañedo Argüelles et al., 2015; Chao, Chazdon, Colwell, & Shen, 2005). Both taxonomic and functional community dissimilarities were then plotted against the duration since flow resumption.

All statistical analyses were carried out using R software (R Development Core Team, 2017) and functions in the package Vegan (Oksanen et al., 2017) and CommEcol (Melo, 2017). Graphics were produced by ggplot2 package for R (Wickham, 2009). For all tests, $p < 0.05$ indicated statistical significance.

3 | RESULTS

3.1 | Dry riverbed environmental characteristics

The average temperature at the surface of the riverbed sediments was $15 \pm 5.6^\circ\text{C}$, while the average sediment moisture during dry phases was $11.9 \pm 8.9\%$ (Table 2). The mean $\pm$ SD grain size (in mm) of whole dataset of 25th, 50th, and 75th percentiles was 3.1 ± 3.3 mm, 19.4 ± 10.1 mm, and 47.9 ± 15.5 mm, respectively, while mean proportion (%) of small particles < 1 mm was $29.0 \pm 10.3\%$ (Table 2).

3.2 | Characteristics of the sampled communities

Taxonomic and functional richness, and total abundance were lowest in DRY, followed by POST and highest in PRE samples (Figure 2a–c and Table S2). DRY samples harboured, on average, fewer species (22 ± 11) than PRE samples and POST samples (60 ± 15 and 45 ± 14 , respectively; $F_{2,51} = 36.7$, $p < 0.001$) but the loss of functional traits was less pronounced (10 ± 3 in DRY, 16 ± 2 in PRE, and 14 ± 2 in POST; $F_{2,51} = 29.8$, $p < 0.001$). The same pattern followed the log-transformed total abundance being lowest in DRY (5.2 ± 0.9) and higher in PRE and POST (8.3 ± 0.7 and 6.8 ± 0.9 , respectively; $F_{2,51} = 59.6$, $p < 0.001$).

PRE samples were dominated in abundance by Diptera 44% (Chironomidae prevailed), and Ephemeroptera, Oligochaeta, and Plecoptera represented $>10\%$. On average, 22 taxa were found alive in DRY samples (Table S2 and Video S1), representing a proportion of 83% of all recorded taxa (the remaining being collected dead). In terms of abundance, a total of 69% of all organisms were found alive (i.e. 4,961 organisms alive and 2,228 dead). DRY samples were dominated by Oligochaeta (25%), Coleoptera, (23%), and Diptera (20%), and similarly, POST samples were dominated by Oligochaeta (24%), Diptera (18%), and Coleoptera (14%) accompanied with Plecoptera (16%; Table S2).

3.3 | Influence of dry and flowing phases on taxonomic and functional richness

In DRY samples, taxonomic and functional richness decreased exponentially with increasing dry phase duration ($F_{1,16} = 16.6$, $p < 0.001$ for taxonomic richness and $F_{1,16} = 23.8$, $p < 0.001$ for functional richness, respectively). In general, doubling the duration of the dry phase (an increase of dry phase duration by 100%) resulted on average in the loss of 3.15 taxa and 1.4 functional units (Figure 3). The decrease was lower for functional richness than for taxonomic richness (TR; $F_{1,32} = 7.97$, $p = 0.008$).

After flow resumption, both taxonomic and functional richness increased exponentially with duration of the flowing phase (Figure 4, $F_{1,16} = 20.1$, $p < 0.001$ for taxonomic and $F_{1,16} = 7.1$, $p = 0.017$ for functional richness, respectively). The increase in taxonomic richness was higher than that in functional richness (FR; $F_{1,32} = 23.95$, $p < 0.001$). In general, doubling the duration of the flowing phase (an increase of flowing phase by 100%) resulted in an increase of

15.4 taxa and 1.8 functional units in POST samples in average. Even after several months of rewetting, the number of species in POST samples was still largely below that of PRE samples ($F_{1,34} = 10.2$, $p < 0.001$, Figure 2a, compare with Figure S2 for perennial sites where the number of taxa is more stable).

3.4 | Nestedness and turnover components of taxonomic and functional β -diversity

The total β -diversity was higher for taxonomic than for functional composition for all sample types (PRE, DRY and POST, Kruskal–Wallis $p < 0.001$). Overall, total β -diversity was highest in DRY samples and lowest in PRE samples (Figure 5). For both taxonomic and functional β -diversity, the turnover component was dominant across all sample types, ranging from 5 to 100% of the total β -diversity with a mean of 82.7% (Figure 5). In contrast, the nestedness component was marginal for each sample type (Figure 5).

3.5 | Contribution of communities surviving during dry phases to benthic community recovery

In terms of number of shared taxa, the contribution of DRY samples to POST samples was as high as 43% just 10 days upon rewetting (with 12 taxa in common), but gradually decreased with the duration of flowing period (Figure 6a, $F_{1,12} = 20.1$, $R^2 = 0.59$, $p < 0.001$). After 60 days of flowing conditions, this contribution was <15% with only nine taxa in common (Figure 6a). With respect to community composition, the similarity between DRY and POST samples was about 60% during the early phases of flow resumption (i.e. 10 days) and also decreased gradually with the duration of flowing period (Figure 6b, $F_{1,12} = 6.2$, $R^2 = 0.28$, $p = 0.029$). After 60 days of flowing conditions, this contribution was <25% (Figure 6b).

Rather similar patterns were found for functional diversity. In terms of number of shared traits, the contribution of DRY samples to POST samples was about 75% 10 days upon rewetting (with 9 trait units in

TABLE 2 Environmental characteristics of the studied dry riverbeds

Variables	Mean	SD	Min.	Max.
Temperature on the dry riverbed (°C)				
Total	15.0	5.6	-11.1	56.6
Day	16.1	5.8	-10.2	56.6
Night	13.3	4.8	-11.1	29.3
Sediment moisture (% water content)	11.9	8.9	0.4	50.0
Grain size (mm)				
25% percentile	3.1	3.3	0.1	13.3
50% percentile	19.4	10.1	1.0	40.2
75% percentile	47.9	15.5	18.6	87.9
Particles < 1 mm (%)	29.0	10.3	13.4	55.2

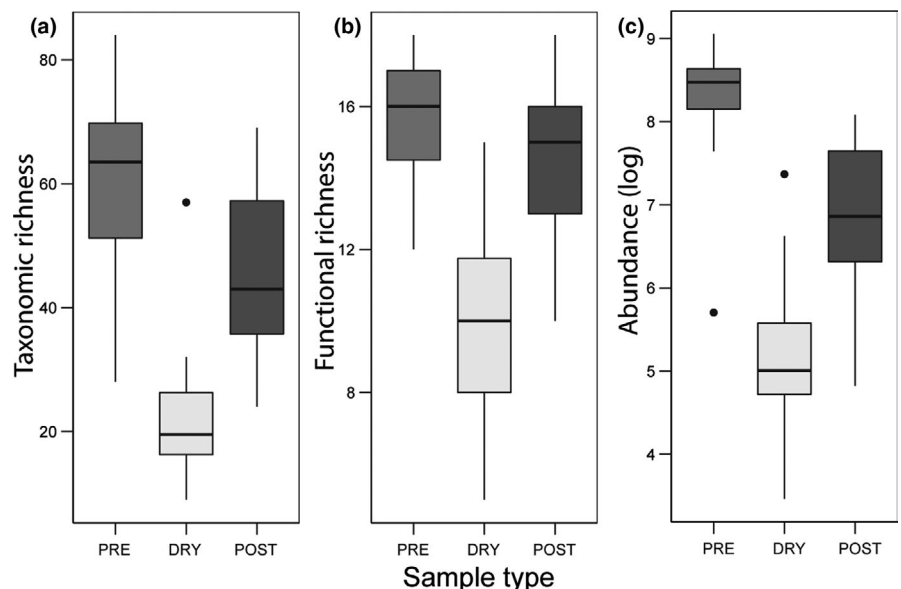


FIGURE 2 Differences in (a) taxonomic and (b) functional richness, and (c) abundance of invertebrates among PRE, DRY and POST samples

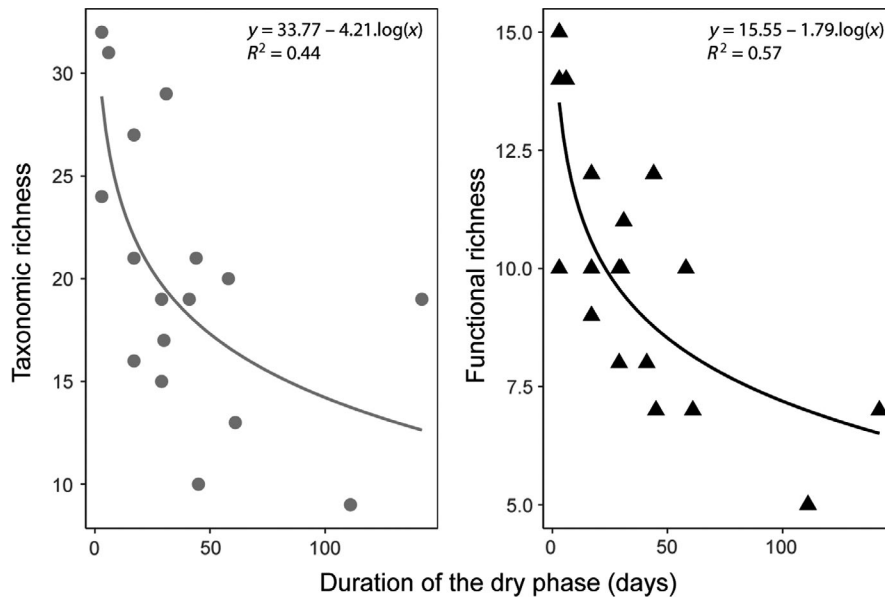


FIGURE 3 Relationships between taxonomic (grey circles) and functional (black triangles) richness of dry riverbed invertebrate communities and the duration of dry phases

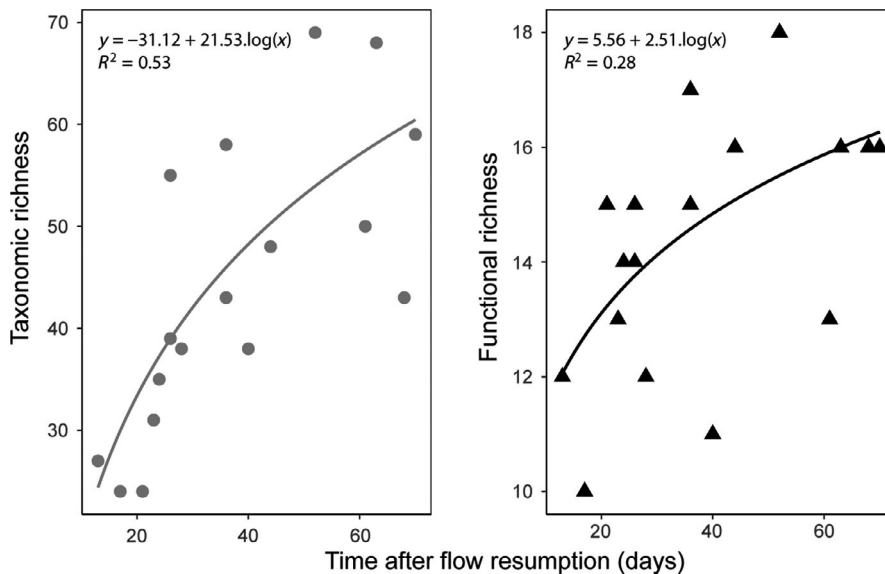


FIGURE 4 Relationships between taxonomic (grey circles) and functional (black triangles) richness of invertebrate communities during flowing phases and the time after flow resumption

common), but gradually decreased with the duration of flowing period (Figure 6c, $F_{1,11} = 7.8$, $R^2 = 0.36$, $p = 0.020$). After 60 days of flowing conditions, this contribution was c. 57% with only seven traits in common (Figure 6c). With respect to functional composition, the similarity between DRY and POST samples was c. 80% during the early phases of flow resumption (i.e. 10 days) but no significant decrease with the duration of flowing period was found (Figure 6d). After 60 days of flowing conditions, this contribution was approximately 76% (Figure 6d).

4 | DISCUSSION

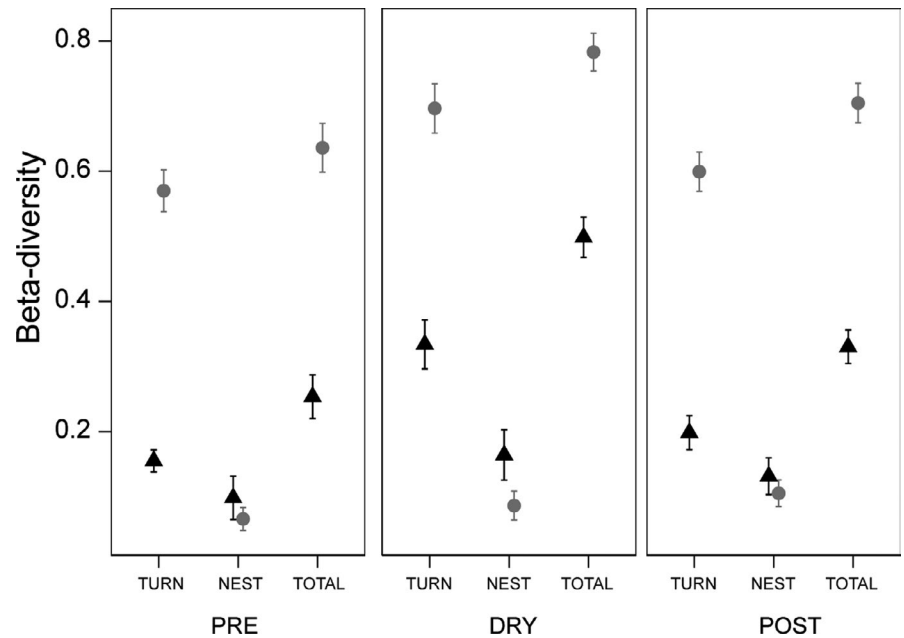
We explored the ability of aquatic invertebrates without resistance adaptations from continental humid climates to cope with anthropogenically-induced (i.e. water abstraction, land-use alterations, climate change) flow intermittence. Contrary to what is reported from mesic

climates, where natural flow intermittence promoted a development of physiological resistance to desiccation (Strachan et al., 2015, 2016; Stubbington & Datry, 2013), and owing to the high relative air humidity, riparian cover, and the short duration of drying events, we expected taxonomically and functionally rich communities to persist in the dry streambed during phases with no surface water. This prediction was supported, showing how stream invertebrate communities may survive through periods of anthropogenically-induced flow intermittence, in continental regions of Europe where increasing numbers of streams are affected by anthropogenic flow intermittence.

4.1 | Communities of invertebrates on dry riverbed sediments

On average, 22 taxa (83% of the total number of taxa collected during the dry periods) were found alive in the sampled sediments,

FIGURE 5 Beta-diversity components (turnover, TURN; nestedness, NEST; and total) for functional and taxonomic richness for PRE, DRY, and POST samples. Mean and 95% confidence interval are plotted for each sample type separately



which is as high as previous studies have found emerging from dry sediments that were artificially rewetted for 3 weeks (e.g. 22 taxa in Boulton, 1989; 24 taxa in Datry et al., 2012) and which included eggs and other desiccation-resistant forms. The pool of taxa surviving moist conditions during the dry phase in the 10 riverbeds of this study is richer than previously expected (Datry et al., 2012; Strachan et al., 2015, 2016; Stubbington, Gunn, Little, Worrall, & Wood, 2016), indicating that it could be an important source of resilience in continental areas, as suggested previously (Strachan et al., 2016). On average, only 30% of organisms were found dead during the dry period, indicating that their large proportion was able to cope with the temporary lack of surface water. Dry riverbed communities were dominated by Oligochaeta (mostly semi-terrestrial Lumbricidae and Enchytraeidae), Coleoptera (Hydraenidae and Elmidae), and Diptera (mostly Chironomidae), which are frequently reported from rewetted sediments (Datry et al., 2012; Stubbington & Datry, 2013). Their proportions were much higher in dry riverbeds compared to samples collected during flowing phases, which suggests that they are tolerant to drying. In contrast, several taxa found abundantly during the flowing phases such as Plecoptera (Nemouridae, Taeniopterygidae, and Capniidae), Ephemeroptera (Leptophlebiidae, Baetidae), and Amphipoda (*Gammarus fossarum*) were only occasionally found alive in dry sediment samples for short drying durations, indicating a low tolerance of these specific stages to drying. In terms of functional aspects, the number of functional units in DRY samples (10 in average) were half of that in POST (14) or PRE (16) samples, yet DRY samples seemed to contribute substantially to the recovery of functional diversity of stream communities upon rewetting, with up to 63% of shared traits on average. However, this contribution also varied with duration of drying and time since rewetting.

This high survival of organisms on these streams during the dry phase can be explained by the mild environmental conditions

prevailing in the dry riverbeds. During sampling, temperature at the surface of the dry sediments was 16°C on average during daylight, while mean sediment moisture stay 11.9% relatively high, even after 4 months of drying. In packs of leaf litter deposited over dry sediments, moisture content could be as high as 50% and mean air moisture ranged from 80% at the surface to 96% in the top 5 cm below the surface (Havelka, 2018). These conditions are much more benign than those reported from Mediterranean (Steward et al., 2011), oceanic temperate (Datry et al., 2012), or arid (Stanley, Buschman, Boulton, Grimm, & Fisher, 1994) dry riverbeds, where temperature can exceed 60°C at the bed surface for the most of the daylight. The high (90–100%) riparian cover at most sites probably protected dry riverbeds from the high temperatures and decrease of air humidity. This emphasises a need to couple the management of rivers and their riparian zones, which is rarely done in practice (Storey & Quinn, 2013).

4.2 | Effects of drying duration on invertebrate communities

As predicted, both taxonomic and functional richness of dry riverbed communities decreased with increasing duration of the dry phases. While the duration of the dry phase was shown previously to be a key determinant of invertebrate seedbank composition (Larned, Datry, & Robinson, 2007; Storey & Quinn, 2013; Stubbington & Datry, 2013) and globally of stream community persistence (Bogan et al., 2013; Datry, 2012; Leigh & Datry, 2017) in arid and Mediterranean intermittent streams, our results indicate it also controls the number of taxa and traits able to survive throughout the dry phases in continental regions. Several studies reported decreased invertebrate richness with increasing dry period duration when analysing the invertebrate seedbank (Datry et al., 2012; Larned et al., 2007; Stubbington & Datry, 2013), that is, the pool of species producing

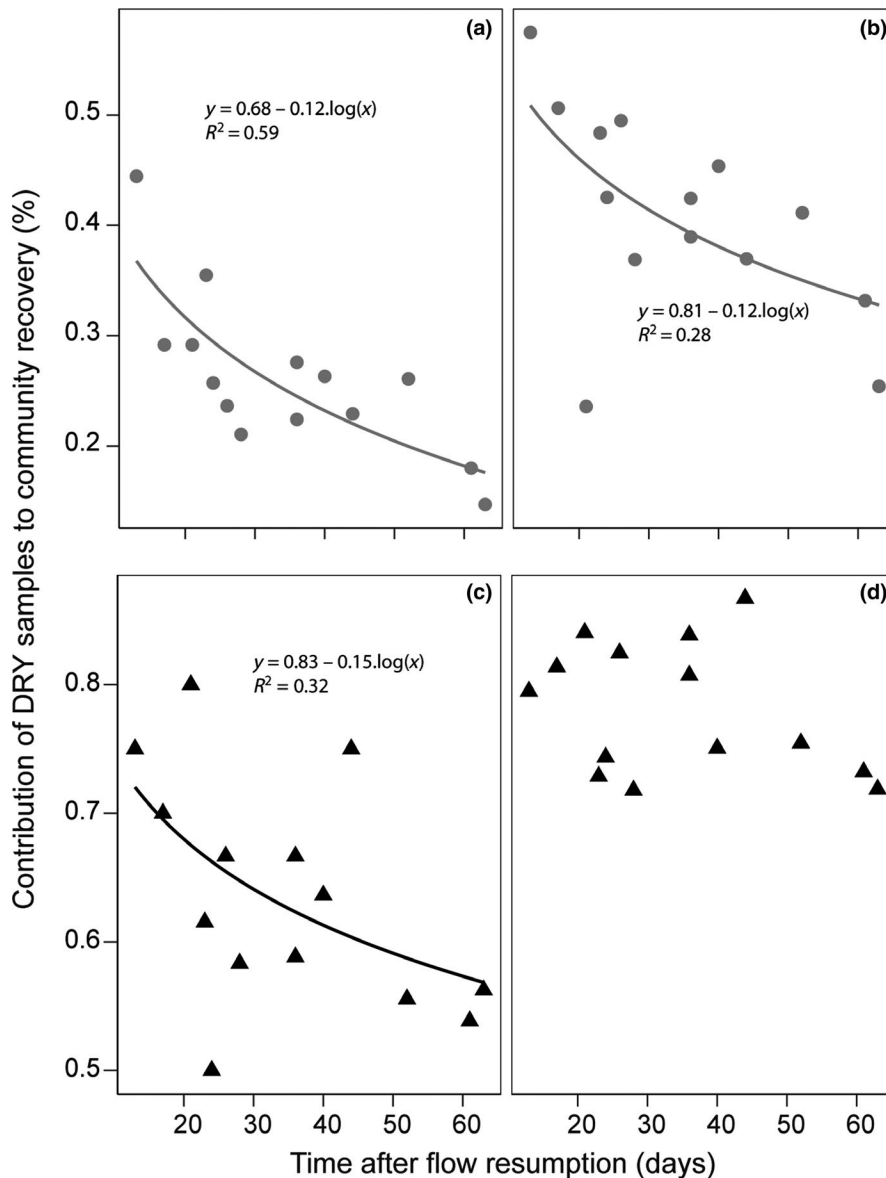


FIGURE 6 Contribution of DRY sample communities to benthic community recovery as % in terms of (a) shared taxa, (b) community composition, (c) shared biological traits, and (d) functional composition

desiccation-resistant forms to cope with dry events. However, to our knowledge, this is the first time that such relationship is documented for the invertebrates surviving as active forms (i.e. not as resistant forms) in dry riverbeds in continental streams.

Contrary to what has been reported for the invertebrate seedbank (Larned et al., 2007; Stubbington & Datry, 2013), the decrease of taxonomic richness (and functional richness) with increasing dry period duration was not linear but exponential. This could be due to the high number of taxa collected here in the first days of the dry phase (on average c. 30 taxa), which then exponentially decreased, while for the invertebrate seedbank, the richness of the pool of resistant species was always <30 and generally <20 taxa per sample. This indicates that there is a pool of invertebrates that cannot survive long without surface water, inducing a sharp decrease during the first 3–4 weeks of dry period, beyond which the decrease became more linear probably because the remaining taxa are the most tolerant to desiccation. Accordingly, drying acts as a ramp disturbance (sensu

Lake, 2000) for the organisms that are not able to escape to perennial refuges (i.e. perennial reaches, adjacent ponds and wetlands, underneath hyporheic zone). However, even after 100 days of drying, c. 20 taxa were still found alive indicating they can tolerate very long dry events. These taxa were Oligochaeta (earthworm *Eiseniella tetraedra* and Enchytraeidae as *Henlea/Fridericia* spp.), Nematoda and Diptera (Ceratopogonidae- *Bezzia* sp., Stratiomyidae- *Oxycera* sp. and Chironomidae-*Chaetocladius piger*-Gr, *Smittia* sp.) which are mostly semi-terrestrial forms and common inhabitants of intermittent rivers worldwide and are also considered as desiccation resistant (Arscott, Larned, Scarsbrook, & Lambert, 2010; Bogan et al., 2013, 2017; Datry, Bonada, & Boulton, 2017; Datry, Vander Vorste, et al., 2017). Our results confirm these previous findings but suggest that this be due to the tolerance of dry conditions, without necessarily requiring the use of desiccation-resistant stages such as eggs, dormant, or quiescent stages (Strachan et al., 2015; Stubbington et al., 2017). Documenting such quantitative relationships will help

predict the effects of increasing stream drying in the CR (Benčoková et al., 2011; Van Lanen et al., 2016; Zahrádková et al., 2015) and potentially in other places with similar climates, as a response to global change and guiding conservation and adaptation strategies.

Both taxonomic and to a lesser extent, functional β -diversity of the communities from dry riverbeds was dominated by turnover, not by nestedness. This pattern was probably due to (1) the high stochasticity of drying in streams from the CR; (2) the lack of adaptation of communities to drying as opposed to what is general observed in more arid climates; and (3) the relatively mild conditions prevailing during dry phases as discussed above. These three factors alone or combined could explain the removal of taxa from the communities with increasing dry phase duration did not follow a consistent pattern leading to nestedness, as often reported along flow intermittence gradients (Datry et al., 2014). In contrast, site-specific responses might have induced turnover, i.e. the taxa responded to drying differently across the studied sites. This context dependency could be due to contrasted local habitat conditions, the variability in biotic interactions (such as the presence of predators and scavengers in the streams or adjacent riparian zones, Steward et al., 2011) and/or other variables not measured in this study. This outcome implies that it might be challenging to predict the effects of flow intermittence in streams recently prone to drying and calls for comparing the responses of communities to natural and artificial flow intermittence.

4.3 | Recovery upon rewetting and community resilience

Although limited in terms of number of taxa, communities collected from dry sediments seem to have an important role in the recovery of benthic communities upon rewetting. About 2 weeks after rewetting, almost half of benthic taxonomic richness was composed of taxa also collected in the dry sediments—this was also true in terms of community composition (i.e. based on the relative abundance of each taxon in the community). This proportion is substantial and exceeds by far the contribution of the invertebrate seedbank found previously in temperate, arid, and neotropical environments (Datry, Bonada, & Boulton, 2017; Datry, Vander Vorste, et al., 2017; Hay, Jenkins, & Kingsford, 2018). The contribution to recovery in terms of functional diversity, based on the traits coded here, was very high. Besides, this contribution did not increase with flow duration as the taxonomic contribution did, indicating that the pool of species on dry sediments allowed a very quick and constant functional recovery. However, additional work is needed to better understand how drying influences the functional traits of river communities (Aspin et al., 2019; Cid et al., 2016; Vander Vorste et al., 2016). Aquatic invertebrates alive in dry riverbeds can thus be a significant source of resilience for benthic communities in these continental streams, despite the intense predation pressure to which dry riverbeds are usually prone (Corti & Datry, 2016; McCluney & Sabo, 2016; Steward et al., 2011).

After 1.5 months of flow resumption, only 20% of the taxa from benthic communities were in common with dry riverbed communities,

reflecting the fast ecological successions documented from IRES (Bogan et al., 2013; Leigh et al., 2016). Taxa from perennial reaches (Arscott et al., 2010; Bogan et al., 2013), underlying hyporheic zones (Datry, 2012), and other refuges located in the catchment colonise gradually previously dry reaches, supporting our initial expectation and explaining why the contribution of dry riverbed communities decreased with time. Such increase in taxonomic and functional richness with time since flow resumption is a common feature of IRES—this study indicates that the initial phases of community recovery are largely driven by the pool of invertebrates surviving over the dry sediments. However, compared with Mediterranean regions (Bonada et al., 2007; Vander Vorste et al., 2016), the recovery seems to be much slower in these continental streams as even 3 months following rewetting, communities were still not fully recovered. Such patterns, often found in intermittent rivers (Datry, 2012; Datry et al., 2014; Datry, Moya, Zubieta, & Oberdorff, 2016; Leigh et al., 2016) were confirmed by the fact that taxonomic richness from 8 adjacent perennial sites showed no differences between pre- and post-drying sampling periods. While this probably reflects the few adaptations shown by species to cope with drying for periods as long as in the Mediterranean region, it provides evidence supporting the need to conserve and protect dry riverbed communities from anthropogenic activities (Steward et al., 2011) to promote community and ecosystem resilience in stream exposed to global change and increased stream network fragmentation by drying.

5 | CONCLUSIONS

Our results indicate that a large pool of invertebrate species can cope with drying without requiring specific adaptations, at least when benign conditions prevail in streambeds during the dry phase. In turn, we quantified for the very first time how this pool of taxa could contribute to community recovery upon rewetting in these forested intermittent streams. In the context of climate change and increasing stream drying, our results show the need for joint management of drying streams and their adjacent riparian zones to promote benign environmental condition allow aquatic invertebrates to survive for weeks or months without surface water. Riparian forest management along intermittent streams from continental climates, like these from the CR, could help conserve the value of dry streambeds as source of resilience for riverine communities.

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SUPPORTING INFORMATION

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